

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 1.4 CONGRUENCE: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Use what you learned in all 8 lessons to answer the driving question: How does the mathematics of reflection explain why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy, and how can we use that same idea to prove that two shapes are truly identical? Write your answers in the space under each prompt. Use full sentences, show your working, and use these terms where they fit: line of reflection, perpendicular distance, image, congruence, and $y = mx + c$. Draw neat labelled diagrams where they help. Use the evidence you collected in class, such as your coordinate tables and the measurements you made, to support every claim.

Part 1: The Rule the Water Follows

At dawn near Kisumu, a fisherman's boat casts a mirror image on the still water of Lake Victoria. The image is flipped, but it looks exactly the same size as the boat. (a) Treat the waterline as the x-axis. Describe the rule that the water surface applies to every point of the boat, and write it as a coordinate rule. (b) Describe what you found when you measured the perpendicular distance from a vertex and from its image to the line of reflection. (c) Explain why this means the reflection is neither bigger nor smaller than the boat.

Write your answer here:

Part 2: Different Mirror Lines

The waterline will not always be on the x-axis. (a) Triangle ABC has A(1, 3). Find the image of A when it is reflected across each of these lines: $y = 2$, $x = 3$, $y = x$, and $y = -x$. Show how you worked out each one. (b) Look at the four rules you used. What stays the same in all four reflections, even though the mirror line changes direction?

Write your answer here:

Part 3: The Hidden Waterline

One morning, mist hides the waterline. You can see a point on the boat, $P(2, 5)$, and its mirror image $P'(6, 5)$. (a) Explain how you can use only these two points to find the line of reflection. (b) Find the equation of the line of reflection. (c) Explain why your method works, using the idea of perpendicular distance.

Write your answer here:

Part 4: Proving Two Shapes Are Identical

Triangle PQR on the boat is reflected in the waterline to give triangle P'Q'R'. (a) What does it mean mathematically for two shapes to be congruent? (b) Name the congruence condition you would use to prove that PQR is congruent to P'Q'R', and write out the steps of the proof. (c) Write the congruence statement with the vertices in the correct order. (d) List the other four congruence conditions.

Write your answer here:

Part 5: Using the Idea in Real Life

Now answer the full driving question and apply it. (a) Write a clear explanation of why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy, and how the same idea proves two shapes are identical. (b) Give one real-life Kenyan example from the unit where reflection or congruence is used, and explain how the mathematics helps.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining reflection as the rule the water follows	Clearly states that every point and its image are the same perpendicular distance from the line of reflection, links this to the same size of the image, and gives correct coordinate rules.	States the equal-distance idea and gives the rule for the x-axis, but the link to the same size is not fully explained.	Says the image looks the same but gives no rule or no mention of equal perpendicular distance.
Accuracy of reflection calculations and the mirror line equation	All images are correct for every mirror line and the mirror line is found correctly as a perpendicular bisector, with working shown.	Most images are correct, and the mirror line is found with small errors in working.	Several images are wrong or the mirror line is not found.
Congruence proof	Names SSS, lists all three pairs of equal sides, writes the congruence statement in correct vertex order, and names the other four conditions.	Names the condition and writes the statement but leaves out a step or a condition.	Does not name a correct condition or does not give a congruence statement.

Use of evidence and mathematical vocabulary	Uses line of reflection, perpendicular distance, image, congruence and $y = mx + c$ correctly, and supports claims with values from class work.	Uses most of the terms correctly, with some claims given without evidence.	Uses few terms or uses them incorrectly, with little evidence.
Connecting to the phenomenon and real life	Gives a complete explanation of the Lake Victoria mirror image and a correct Kenyan application with clear reasoning.	Explains the mirror image and gives an application, but the reasoning is incomplete.	Gives a partial explanation, or an application with no reasoning.

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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